Risk Management in Software Engineering

RCS 305 ? Software Engineering | Riara University Revision Resource

1. Proactive vs. Reactive Risk Strategies

Proactive risk management identifies potential failure modes before project inception, formalizing mitigation plans. Reactive strategies ('fire fighting') address crises only after damage has materialized.

2. The Systematic Risk Management Cycle

? 1. Identification: Cataloging risks across technical, personnel, project, and environmental categories.
? 2. Analysis & Exposure Calculation: Determining Probability (P) and Impact (I). Risk Exposure (RE) = Probability x Impact.
? 3. Prioritization: Matrix ranking (Critical, High, Medium, Low) to allocate defensive resources.
? 4. Mitigation, Monitoring & Management (RMMM Plan): Drafting contingency protocols and tracking leading indicators throughout execution.

3. Common Software Risk Categories

Requirements volatility, key personnel turnover, technology obsolescence, third-party vendor default, and schedule compression.

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